Project Dataset

Dataset: Brazilian Olist E-Commerce Dataset

Data Integration and Data Quality

The datasets were integrated using common customer, order, product, and payment identifiers. Missing values, duplicate records, and inconsistent data were checked and handled during preparation.

EDA and Business Findings

Sales patterns, customer behavior, product categories, payment methods, and delivery performance were analyzed to identify useful business insights.

Feature Engineering

Features such as recency, frequency, monetary value, delivery delay, and review score were prepared to support customer segmentation and predictive modeling.

Customer Segmentation

K-Means clustering was applied using scaled customer features. Elbow and Silhouette methods were used to compare cluster numbers and identify meaningful customer groups.

Predictive Modeling and Evaluation

Multiple classification models were trained and evaluated using suitable metrics, including accuracy, precision, recall, F1-score, and ROC-AUC. The best model was selected based on its evaluation results.

Model Improvement and Tuning

Cross-validation and hyperparameter tuning were used to improve model performance. Training and testing results were compared to check for overfitting.

Sales Time Series Analysis and Forecasting

Monthly sales were analyzed to identify trends. A forecasting model was evaluated using a reserved test period and an appropriate error metric.

Data Degradation

Missing values, inconsistent records, delivery delays, and incomplete customer information can reduce data quality and affect model reliability.

Data Equity

Customer and order data may not represent every customer group equally. Model results should be monitored for bias, and important business decisions should include human review.

Business Questions and Answers

1. What was the most difficult data integration problem?
Connecting multiple datasets through common identifiers while handling missing and inconsistent records.

2. Which engineered feature added the most business value?
RFM features help identify customer purchasing behavior and support targeted marketing.

3. Which customer segment is the most valuable?
The segment with the highest monetary value and stronger purchase frequency is the most valuable.

4. Which predictive model performed best?
The model with the strongest evaluation results was selected based on the notebook's model comparison.

5. How was overfitting checked?
Training and testing performance were compared, along with cross-validation results.

6. What business patterns were identified?
Sales trends, customer purchasing behavior, delivery performance, and review patterns were analyzed.

7. What does the monthly forecast suggest?
The forecast supports sales planning, inventory management, and resource allocation.

8. What is the main data limitation?
Missing information, historical data limitations, and possible customer-group imbalance may affect the conclusions.

Conclusion

The project combines data preparation, customer segmentation, predictive modeling, and sales forecasting to support data-driven business decisions. The results can guide marketing, customer retention, and operational planning.